

Stage 2 - Data Preparation and Scaling

Project: Customer Churn Analysis for Telecommunication Company

Team: WILDA Team 6

1. Objective

This stage transforms the supplied customer churn dataset into a consistent, reproducible format suitable for modelling and clustering. The workflow includes data-quality checks, encoding, target separation, train-test splitting and appropriate scaling.

2. Source Dataset Review

Measure	Result
Records	7,043
Columns	10
Target variable	Churn
Missing values	0 across all columns
Churn = No	5,174
Churn = Yes	1,869
Exact duplicate-looking rows	302 retained

No missing values were found, so imputation was not required. The 302 exact duplicate-looking rows were retained because the reduced dataset has no customer ID; matching values across the limited fields do not prove that the rows represent the same customer.

3. Feature and Target Treatment

The target variable **Churn** is encoded as No = 0 and Yes = 1 and kept separate from the predictors. This prevents the known outcome from being used as an input feature.

Variable type	Variables	Treatment
Nominal categorical	gender, Dependents, PhoneService, MultipleLines, InternetService, Contract	One-hot encoding
Binary numeric	SeniorCitizen	Retained as 0/1; not scaled
Continuous numeric	tenure, MonthlyCharges	StandardScaler in model-ready train/test predictors
Target	Churn	No = 0, Yes = 1; kept separate from X

4. Categorical Encoding

One-hot encoding is used for nominal variables because categories such as contract type and internet service have no meaningful numeric order. This avoids introducing an artificial ranking.

```
OneHotEncoder(handle_unknown="ignore", sparse_output=False)
```

5. Training and Testing Split

An 80/20 split is used with **random_state=42** for reproducibility. Stratification by Churn preserves the class proportions in the training and testing subsets.

Set	Records	Churn No (0)	Churn Yes (1)
Training	5,634	4,139	1,495
Testing	1,409	1,035	374

```
train_test_split(X, y, test_size=0.20, random_state=42, stratify=y)
```

6. Scaling Technique

StandardScaler is applied to the continuous variables **tenure** and **MonthlyCharges**. The binary SeniorCitizen indicator remains 0/1 because scaling a binary flag is not necessary for this preparation. StandardScaler centres each continuous

feature using the training-set mean and divides by the training-set standard deviation.

The preprocessing object is fitted only on the training set and then used to transform the test set. This avoids data leakage because the test-set values do not influence the learned scaling parameters.

```
X_train_processed = preprocessor.fit_transform(X_train)
X_test_processed  = preprocessor.transform(X_test)
```

7. Output Files

File	Purpose
customer_churn_preprocessed.csv	Full encoded dataset with interpretable unscaled numeric values and the Churn target.
X_train.csv / X_test.csv	Model-ready predictors; nominal categories encoded and continuous features scaled.
y_train.csv / y_test.csv	Separate Churn targets.
split_summary.txt	Documents split size and target composition.
data_preparation.ipynb	Documented executable notebook.
data_preparation.py	Reproducible preprocessing script.
preprocessing_pipeline.joblib	Fitted preprocessing object for reproducibility.

8. Validation and Conclusion

Validation confirms that all 7,043 records are accounted for, train and test predictors have the same 16-column structure, no invalid numeric values were introduced, and the scaled continuous training features have means approximately 0 and standard deviations approximately 1. The outputs are ready for the Stage 2 modelling and clustering work.